

Exercises

Exercise 1: Power-flow sign convention Exercise

and Arts

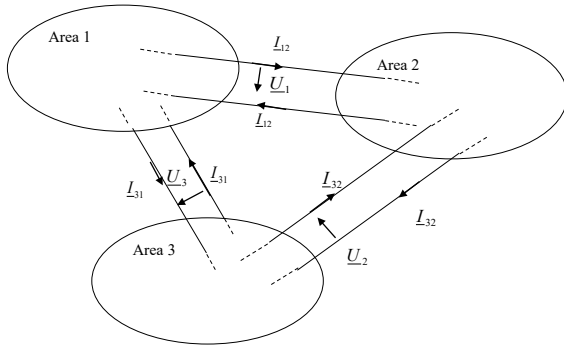
Exercises Electric Power Grids (EPG)

EPG-Exercise 1: Interpretation of numerical values in sign convention

Three areas are connected to each other by single-phase lines as shown in the figure below. The following simultaneous current and voltage values are measured:

Data: $\underline{U}_1 = 76 \angle 4^\circ \text{ kV}$, $\underline{U}_2 = 76 \angle 8^\circ \text{ kV}$, $\underline{U}_3 = 76 \angle -11^\circ \text{ kV}$
 $\underline{I}_{12} = 450 \angle -15^\circ \text{ A}$, $\underline{I}_{31} = 300 \angle 18^\circ \text{ A}$, $\underline{I}_{32} = 500 \angle -11^\circ \text{ A}$

- Which areas provide active power? How much?
- Which areas receive reactive power? How much?



Formulas needed

With the reference directions shown in the exercise:

$$\underline{S} = \underline{U} \underline{I}^*, \quad P = \Re\{\underline{S}\}, \quad Q = \Im\{\underline{S}\}.$$

For phasors,

$$(A \angle \alpha)(B \angle \beta)^* = AB \angle (\alpha - \beta).$$

Positive P or Q means that the area supplies active or reactive power according to the indicated current references; a negative result means that it receives power.

Complete solution

Area 1

Apply the indicated voltage and current references at the two connections of area 1:

$$\begin{aligned} \underline{S}_1 &= \underline{U}_1 \underline{I}_{12}^* - \underline{U}_3 \underline{I}_{31}^* \\ &= 76 \angle 4^\circ \cdot 450 \angle 15^\circ \\ &\quad - 76 \angle (-11^\circ) \cdot 300 \angle (-18^\circ) \text{ MVA}/1000 \\ &= 34.2 \angle 19^\circ - 22.8 \angle (-29^\circ) \\ &= \boxed{12.40 + j22.19 \text{ MVA}}. \end{aligned}$$

Therefore area 1 supplies 12.40 MW and 22.19 Mvar.

Area 2

$$\begin{aligned} \underline{S}_2 &= \underline{U}_2 \underline{I}_{32}^* - \underline{U}_1 \underline{I}_{12}^* \\ &= 76 \angle 8^\circ \cdot 500 \angle 11^\circ \\ &\quad - 76 \angle 4^\circ \cdot 450 \angle 15^\circ \text{ MVA}/1000 \\ &= 38.0 \angle 19^\circ - 34.2 \angle 19^\circ \\ &= \boxed{3.59 + j1.24 \text{ MVA}}. \end{aligned}$$

Area 2 supplies 3.59 MW and 1.24 Mvar.

Area 3

$$\begin{aligned} \underline{S}_3 &= \underline{U}_3 \underline{I}_{31}^* - \underline{U}_2 \underline{I}_{32}^* \\ &= 76 \angle (-11^\circ) \cdot 300 \angle (-18^\circ) \\ &\quad - 76 \angle 8^\circ \cdot 500 \angle 11^\circ \text{ MVA}/1000 \\ &= 22.8 \angle (-29^\circ) - 38.0 \angle 19^\circ \\ &= \boxed{-15.99 - j23.43 \text{ MVA}}. \end{aligned}$$

Area 3 receives 15.99 MW and 23.43 Mvar. The check $\underline{S}_1 + \underline{S}_2 + \underline{S}_3 \simeq 0$ confirms the lossless power balance.

Exercise 2: Merit-order pricing

Exercise

Exercises Electrical Power Grids (EPG)

Page 2

EPG-Exercise 2: Merit order principle for pricing on the spot market

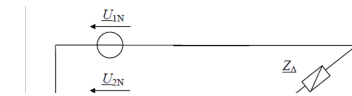
The energy prices are formed on the spot market with the help of a double auction. Sorted list of B1-B6 consumer offers:

B1	B2	B3	B4	B5	B6
100 MWh	200 MWh	250 MWh	250 MWh	150 MWh	150 MWh
0.08 Fr./kWh	0.09 Fr./kWh	0.10 Fr./kWh	0.11 Fr./kWh	0.12 Fr./kWh	0.13 Fr./kWh

Sorted list of offers of electricity sellers S1-S5:

S1	S2	S3	S4	S5
50 MWh	200 MWh	250 MWh	250 MWh	400 MWh
0.07 Fr./kWh	0.075 Fr./kWh	0.085 Fr./kWh	0.095 Fr./kWh	0.15 Fr./kWh

EPG-Exercise 3: Three-phase power



Formulas needed

The supply curve is built by accumulating seller quantities from the lowest to the highest offer price. The demand curve is accumulated from the highest to the lowest willingness to pay. The market-clearing point is their intersection. All accepted offers are settled at the clearing price.

Complete solution

The cumulative supply is

p [CHF/kWh]	.070	.075	.085	.095	.150
E_S [MWh]	50	250	500	750	1150

and the cumulative demand is

p [CHF/kWh]	.130	.120	.110	.100	.090	.080
E_D [MWh]	150	300	550	800	1000	1100

At 0.095 CHF/kWh the sellers offer 750 MWh, while demand is still at least 800 MWh. The next seller asks 0.15 CHF/kWh, which is above the next accepted demand bid. Hence

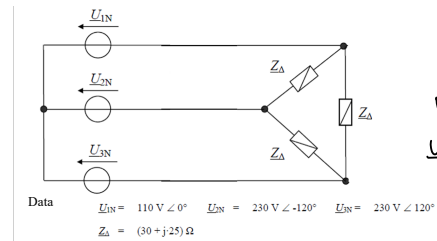
$$p_{\text{clear}} = 0.10 \text{ CHF/kWh}, \quad E_{\text{clear}} = 750 \text{ MWh}.$$

Sellers S1-S4 are fully accepted. Buyers B4-B6 take 550 MWh in total, and B3 is partially accepted for the remaining 200 MWh. S5, B1 and B2 are not accepted.

Exercise 3: Three-phase complex power

Exercise

EPG-Exercise 3: Three-phase power



Calculate the complex power $\underline{S}$ and the power factor λ .

$$U = RI$$

$$U = \underline{Z} \underline{I} \quad \underline{I} = \frac{\underline{U}_\Delta}{\underline{Z}}$$

Dr. Severin Nowak

Hochschule Technik und Architektur Luzern, 6048 Horw, 14.02.2024

Formulas needed

For each delta branch:

$$\underline{U}_{ij} = \underline{U}_{iN} - \underline{U}_{jN}, \quad \underline{I}_{ij} = \frac{\underline{U}_{ij}}{\underline{Z}_\Delta}, \quad \underline{S}_{ij} = \underline{U}_{ij} \underline{I}_{ij}^*$$

The total power and power factor are

$$\underline{S} = \sum \underline{S}_{ij} = P + jQ, \quad \lambda = \frac{P}{|\underline{S}|} = \cos \varphi.$$

Complete solution

The three line-to-line voltages are

$$\underline{U}_{12} = 110 \angle 0^\circ - 230 \angle (-120^\circ) = 300.50 \angle 41.52^\circ \text{ V},$$

$$\underline{U}_{23} = 230 \angle (-120^\circ) - 230 \angle 120^\circ = 398.37 \angle (-90^\circ) \text{ V},$$

$$\underline{U}_{31} = 230 \angle 120^\circ - 110 \angle 0^\circ = 300.50 \angle 138.48^\circ \text{ V}.$$

Since $\underline{Z}_\Delta = 30 + j25 = 39.05 \angle 39.81^\circ \Omega$,

$$\underline{I}_{12} = 7.69 \angle 1.71^\circ \text{ A},$$

$$\underline{I}_{23} = 10.20 \angle (-129.81^\circ) \text{ A},$$

$$\underline{I}_{31} = 7.69 \angle 98.67^\circ \text{ A}.$$

Alternatively, for every branch $\underline{S}_{ij} = |\underline{U}_{ij}|^2 / \underline{Z}_\Delta^*$. Adding the three branch powers gives

$$\underline{S} = \frac{300.50^2 + 398.37^2 + 300.50^2}{30 - j25} = \boxed{6.675 + j5.562 \text{ kVA}}.$$

Thus

$$|\underline{S}| = 8.689 \text{ kVA}, \quad \boxed{\lambda = 6.675 / 8.689 = 0.768}.$$

Because $Q > 0$, the load is inductive and the power factor is lagging.

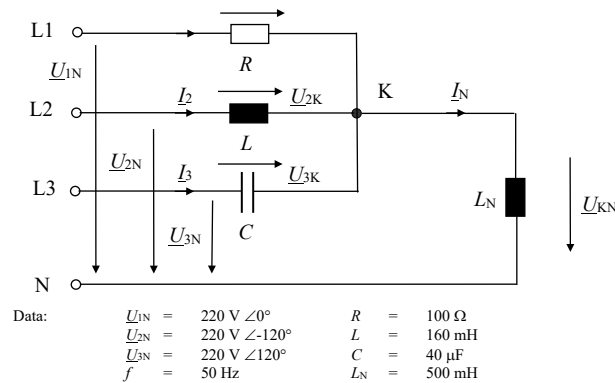
Exercise 4: Unbalanced four-wire system

Exercise

Exercises Electrical Power Grids (EPG)

Page 3

EPG-Exercise 4: 4-wire system



Calculate the currents $\underline{I}_1$, $\underline{I}_2$, $\underline{I}_3$ and $\underline{I}_N$, as well as the voltages $\underline{U}_{1K}$, $\underline{U}_{2K}$, $\underline{U}_{3K}$ and $\underline{U}_{KN}$.

Formulas needed

Use nodal analysis at the floating load star point K:

$$\underline{U}_{KN} = \frac{Y_R \underline{U}_{1N} + Y_L \underline{U}_{2N} + Y_C \underline{U}_{3N}}{Y_R + Y_L + Y_C + Y_N}.$$

Then

$$\underline{U}_{iK} = \underline{U}_{iN} - \underline{U}_{KN}, \quad \underline{I}_i = Y_i \underline{U}_{iK}, \quad \underline{I}_N = Y_N \underline{U}_{KN}.$$

Complete solution

At $\omega = 2\pi 50 = 314.16 \text{ rad/s}$:

$$Y_R = \frac{1}{100} = 0.0100 \text{ S},$$

$$Y_L = \frac{1}{j\omega L} = -j0.019894 \text{ S},$$

$$Y_C = j\omega C = j0.012566 \text{ S},$$

$$Y_N = \frac{1}{j\omega L_N} = -j0.006366 \text{ S}.$$

Substitution into the node equation gives

$$\underline{U}_{KN} = -176.97 - j161.74 = \boxed{239.75 \angle (-137.57^\circ) \text{ V}}.$$

The load voltages are

$$\underline{U}_{1K} = 220 \angle 0^\circ - \underline{U}_{KN} = \boxed{428.66 \angle 22.17^\circ \text{ V}},$$

$$\underline{U}_{2K} = 220 \angle (-120^\circ) - \underline{U}_{KN} = \boxed{72.90 \angle (-23.26^\circ) \text{ V}},$$

$$\underline{U}_{3K} = 220 \angle 120^\circ - \underline{U}_{KN} = \boxed{358.57 \angle 79.24^\circ \text{ V}}.$$

Consequently,

$$\underline{I}_1 = Y_R \underline{U}_{1K} = \boxed{4.287 \angle 22.17^\circ \text{ A}},$$

$$\underline{I}_2 = Y_L \underline{U}_{2K} = \boxed{1.450 \angle (-113.26^\circ) \text{ A}},$$

$$\underline{I}_3 = Y_C \underline{U}_{3K} = \boxed{4.506 \angle 169.24^\circ \text{ A}},$$

$$\underline{I}_N = Y_N \underline{U}_{KN} = \boxed{1.526 \angle 132.43^\circ \text{ A}}.$$

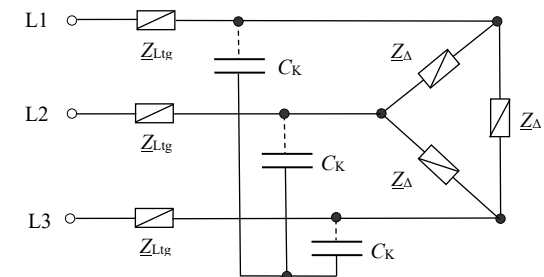
The KCL check is $\underline{I}_1 + \underline{I}_2 + \underline{I}_3 = \underline{I}_N$.

Exercise 5: Balanced three-phase system

Exercises Electrical Power Grids (EPG)

Page 4

EPG-Exercise 5: Calculation of Three-Phase Systems



- Calculate the line currents $\underline{I}$ and voltages $\underline{U}_\Delta$ and the loads $\underline{Z}_\Delta$, if the capacitor bank is not connected.
- Calculate the value of C_k in the capacitor bank for reactive current compensation of the consumers $\underline{Z}_\Delta$.
- Determine the line currents $\underline{I}$ in the compensated case.

Formulas needed

For a balanced system, convert the delta load to star:

$$\underline{Z}_Y = \frac{\underline{Z}_\Delta}{3}, \quad U_{ph} = \frac{U_{LL}}{\sqrt{3}}.$$

One phase can then be solved using

$$\underline{I}_L = \frac{\underline{U}_{ph}}{\underline{Z}_{Ltg} + \underline{Z}_Y}, \quad \underline{U}_Y = \underline{I}_L \underline{Z}_Y, \quad U_\Delta = \sqrt{3} U_Y.$$

For a star-connected capacitor bank,

$$Y_C = j\omega C_K, \quad \Im\{Y_Y + Y_C\} = 0.$$

Complete solution**a) Capacitor bank disconnected**

$$\underline{Z}_Y = \frac{30\angle 60^\circ}{3} = 10\angle 60^\circ = 5 + j8.660 \, \Omega, \quad U_{ph} = 400/\sqrt{3} = 230.94 \, \text{V}.$$

The per-phase total impedance is

$$\underline{Z}_{tot} = (1 + j2) + (5 + j8.660) = 6 + j10.660 = 12.233\angle 60.63^\circ \, \Omega.$$

Taking $\underline{U}_{1N} = 230.94\angle 0^\circ \, \text{V}$,

$$\underline{I}_1 = \frac{230.94\angle 0^\circ}{12.233\angle 60.63^\circ} = 18.879\angle(-60.63^\circ) \, \text{A}.$$

The other line currents are displaced by -120° and $+120^\circ$:

$$\underline{I}_2 = 18.879\angle 179.37^\circ \, \text{A}, \quad \underline{I}_3 = 18.879\angle 59.37^\circ \, \text{A}.$$

The equivalent star-load phase voltage is

$$\underline{U}_Y = \underline{I}_1 \underline{Z}_Y = 188.79\angle(-0.63^\circ) \, \text{V}.$$

Therefore the magnitude across every delta branch is

$$U_\Delta = \sqrt{3} 188.79 = 326.99 \, \text{V}.$$

For example, $\underline{U}_{12,\Delta} = 326.99\angle 29.37^\circ \, \text{V}$; the other two delta voltages are shifted by $\mp 120^\circ$. The load remains $\underline{Z}_\Delta = 30\angle 60^\circ \, \Omega$ per branch.

b) Reactive-current compensation

The star-equivalent load admittance is

$$\underline{Y}_Y = \frac{1}{5 + j8.660} = 0.0500 - j0.086603 \, \text{S}.$$

The capacitor must provide $+j0.086603 \, \text{S}$ per phase:

$$\omega C_K = 0.086603 \implies C_K = \frac{0.086603}{2\pi 50} = 275.7 \, \mu\text{F}$$

per star-connected capacitor.

c) Compensated line current

The compensated parallel load has

$$\underline{Y}_{\text{load,comp}} = 0.0500 \, \text{S}, \quad \underline{Z}_{\text{load,comp}} = 20.0 \, \Omega.$$

Including the line impedance,

$$\underline{Z}_{\text{tot,comp}} = 21 + j2 = 21.095\angle 5.44^\circ \, \Omega.$$

Hence

$$\underline{I}_1 = 10.948\angle(-5.44^\circ) \, \text{A}.$$

Again, $\underline{I}_2$ and $\underline{I}_3$ are shifted by -120° and $+120^\circ$. Compensation reduces the current from 18.88 A to 10.95 A and leaves only the line-reactance phase shift.

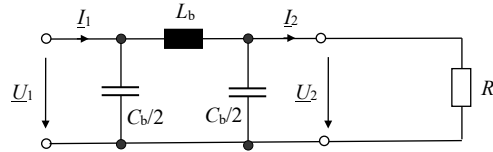
Exercise 6: Surge impedance loading**Exercise**

Exercises Electrical Power Grids (EPG)

Page 5

EPG-Exercise 6: Surge impedance loading

Equivalent line model of a lossless line with load R_L :



Determine the value of R_L so that the lossless line is in optimal operation (i.e. surge impedance loading) with:

$$\begin{aligned} L_b' &= 1.3 \, \text{mH}/(\text{km and phase}) \\ C_b' &= 9.6 \, \text{nF}/(\text{km and phase}) \\ l &= 250 \, \text{km (length of the line)} \end{aligned}$$

EPG-Exercise 7: Transformer 1**Formulas needed**

For a lossless line, the surge or characteristic impedance is

$$Z_c = \sqrt{\frac{L_b'}{C_b'}} = \sqrt{\frac{L_b}{C_b}}.$$

At surge-impedance loading, the inductive and capacitive reactive powers cancel and the resistive load is $R_L = Z_c$.

Complete solution

The total per-phase parameters are

$$L_b = L_b' l = 1.3 \frac{\text{mH}}{\text{km}} \cdot 250 \, \text{km} = 0.325 \, \text{H},$$

$$C_b = C_b' l = 9.6 \frac{\text{nF}}{\text{km}} \cdot 250 \, \text{km} = 2.40 \, \mu\text{F}.$$

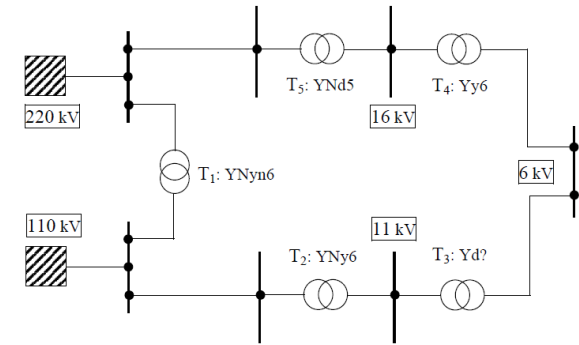
Thus

$$R_L = Z_c = \sqrt{\frac{0.325}{2.40 \cdot 10^{-6}}} = 367.99 \, \Omega \approx 368 \, \Omega.$$

The line length cancels if the distributed values L_b' and C_b' are inserted directly.

Exercise 7: Transformer clock number**Exercise**

$$\begin{aligned} C_b' &= 9.6 \, \text{nF}/(\text{km and phase}) \\ l &= 250 \, \text{km (length of the line)} \end{aligned}$$

EPG-Exercise 7: Transformer 1

Determine the index number of transformer T3.

Dr. Severin Nowak

Hochschule Technik und Architektur Luzern, 6048 Horw, 14.02.2024

Formulas needed

Transformer clock numbers add along a cascade and are evaluated modulo 12:

$$n_{\text{path}} = (n_1 + n_2 + \dots) \bmod 12.$$

Two paths connected to the same buses must produce the same total phase shift.

Complete solution

From the 220-kV bus to the 6-kV bus, the upper path contains T5 and T4:

$$n_{\text{upper}} = (5 + 6) \bmod 12 = 11.$$

The lower path contains T1, T2 and T3:

$$n_{\text{lower}} = (6 + 6 + n_3) \bmod 12 = n_3.$$

The two paths are connected in parallel at their start and end buses, so their phase shifts must be identical:

$$n_3 = 11.$$

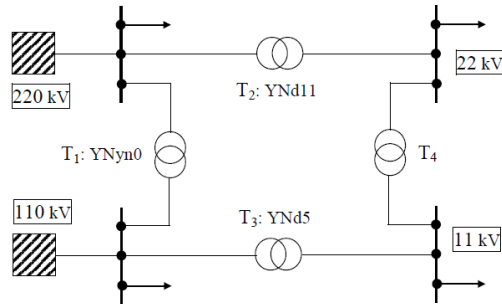
Since the winding configuration is already specified as Yd,

$$\boxed{\text{T3 : Yd11}}.$$

Exercise 8: Suitable transformer group

Exercise

EPG-Exercise 8: Transformer 2



Determine a suitable transformer configuration incl. index number for transformer T4.

Formulas needed

As in Exercise 7, parallel paths must have equal cumulative clock numbers modulo 12.

Complete solution

The lower route from 220 kV to 11 kV contains T1 and T3:

$$n_{\text{lower}} = (0 + 5) \bmod 12 = 5.$$

The upper route contains T2 and T4:

$$n_{\text{upper}} = (11 + n_4) \bmod 12.$$

Equating both routes,

$$11 + n_4 \equiv 5 \pmod{12} \implies n_4 = 6.$$

A delta connection on both sides is suitable for linking the 22-kV and 11-kV buses without introducing a neutral connection. Therefore one suitable choice is

T4 : Dd6.

Exercise 9: Synchronous generator

Exercise

✓ EPG-Exercise 9: Active and reactive power of a synchronous generator

A three-phase synchronous generator (star connected) is built as a nonsalient pole machine. It operates in a grid with constant voltage and frequency (stiff grid).

Data:

Nominal apparent power: $S_N = 400 \text{ MVA}$

Nominal voltage*: $U_N = 20 \text{ kV} / 50 \text{ Hz}$

Synchronous reactance: $X_d = 2 \text{ p.u.}$

In the operation considered, the excitation current is set so that the induced rotor voltage reaches 25 kV (phase voltage). The torque angle is 30° . Determine the active and reactive power of the generator.

What is the electrical behavior of the machine? (capacitive or inductive?)

***Important: if nothing specified, nominal voltage is always the line-to-line voltage.**

Formulas needed

For a cylindrical-rotor synchronous generator with neglected resistance:

$$P_{3\phi} = 3 \frac{VE}{X_d} \sin \delta, \quad Q_{3\phi} = 3 \frac{V}{X_d} (E \cos \delta - V).$$

V and E are phase voltages. Convert the p.u. reactance using

$$Z_B = \frac{U_{LL,N}^2}{S_N}, \quad X_d[\Omega] = X_d[\text{p.u.}] Z_B.$$

Complete solution

The base impedance and synchronous reactance are

$$Z_B = \frac{(20 \text{ kV})^2}{400 \text{ MVA}} = 1 \Omega, \quad X_d = 2 \cdot 1 = 2 \Omega.$$

The grid phase voltage is

$$V = \frac{20}{\sqrt{3}} = 11.547 \text{ kV},$$

whereas the induced rotor phase voltage is $E = 25 \text{ kV}$. Therefore

$$P_{3\phi} = 3 \frac{(11.547 \cdot 10^3)(25 \cdot 10^3)}{2} \sin 30^\circ = \boxed{216.5 \text{ MW}},$$

and

$$Q_{3\phi} = 3 \frac{11.547 \cdot 10^3}{2} (25 \cdot 10^3 \cos 30^\circ - 11.547 \cdot 10^3) = \boxed{175.0 \text{ Mvar}}.$$

The generator is overexcited and supplies positive reactive power to the grid. Seen from the grid, it therefore behaves **capacitively**.

Exercise 10: Interconnected control areas

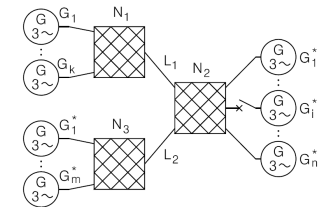
Exercise

EPG-Exercise 10: Interconnected control areas

The interconnected control areas of the grid below are examined (N1, N2, and N3 could, e.g., be three transmission grids of three countries, that are interconnected through the lines L1 and L2, respectively).

At the considered time no power is exchanged on the lines L1 and L2. Due to a fault, a power plant fails in the control area N2. This plant has previously fed 100 MW of power into the grid. The three control areas have the regulating characteristics $K_{N1} = 400 \text{ MW/Hz}$ and $K_{N2} = K_{N3} = 500 \text{ MW/Hz}$.

- What frequency will occur in the control areas after the primary controllers have been activated?
- What power is then exchanged between the areas?
- What input signals ΔP_R are given to the secondary controllers of the three areas?
- Approximately after which time period will the power exchange on the grid lines L1 and L2 be back to normal (i.e., before the disturbance assuming a constant load)?



Formulas needed

After primary control, every synchronous area has the same frequency deviation:

$$\Delta P_i = -K_{Ni} \Delta f, \quad \sum_i \Delta P_i = \Delta P_{\text{loss}}.$$

Using positive tie-line power for export, a convenient secondary-control error is

$$\Delta P_{R,i} = -\Delta P_{\text{tie},i} - K_{Ni} \Delta f.$$

Complete solution

a) Frequency after primary control

The total regulating characteristic is

$$K_{\Sigma} = 400 + 500 + 500 = 1400 \text{ MW/Hz}.$$

For a generation loss of 100 MW,

$$\Delta f = -\frac{100}{1400} = -0.07143 \text{ Hz}, \quad \boxed{f = 49.9286 \text{ Hz}}.$$

b) Power exchange

The primary power increases are

$$\Delta P_{N1} = -400(-0.07143) = 28.57 \text{ MW},$$

$$\Delta P_{N2} = -500(-0.07143) = 35.71 \text{ MW},$$

$$\Delta P_{N3} = -500(-0.07143) = 35.71 \text{ MW}.$$

N2 still lacks $100 - 35.71 = 64.29 \text{ MW}$. It is supplied through the tie lines:

$$\boxed{\begin{matrix} P_{L1} = 28.57 \text{ MW} & (N1 \rightarrow N2) \\ P_{L2} = 35.71 \text{ MW} & (N3 \rightarrow N2) \end{matrix}}.$$

c) Secondary-controller inputs

For N1, $\Delta P_{\text{tie},1} = +28.57 \text{ MW}$:

$$\Delta P_{R,1} = -28.57 - 400(-0.07143) = 0.$$

For N3, $\Delta P_{\text{tie},3} = +35.71 \text{ MW}$:

$$\Delta P_{R,3} = -35.71 - 500(-0.07143) = 0.$$

N2 imports 64.29 MW, so $\Delta P_{\text{tie},2} = -64.29 \text{ MW}$:

$$\Delta P_{R,2} = -(-64.29) - 500(-0.07143) = 100 \text{ MW}.$$

Thus only the disturbed area receives a sustained restoration request:

$$\boxed{\Delta P_{R,1} = 0, \quad \Delta P_{R,2} = 100 \text{ MW}, \quad \Delta P_{R,3} = 0}.$$

The sign at a controller output may be reversed depending on its block diagram; physically N2 must increase generation by 100 MW.

d) Restoration time

Secondary control restores the scheduled exchanges and nominal frequency. With the usual reserve-restoration timescale, this occurs after approximately

$$\boxed{15 \text{ minutes}}.$$

Exercise 11: Bus-admittance matrix

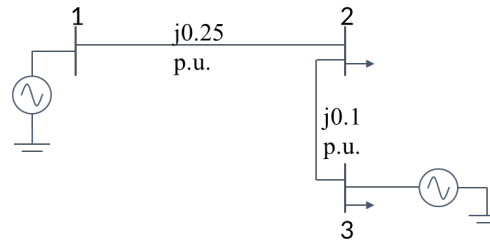
Exercise

Exercises Electrical Power Grids (EPG)

Page 8

✓ EPG-Exercise 11: Ybus

Form the Ybus of the following system:



✓ EPG-Exercise 12: Two-node system

For the following 2-node system, program a simple iterative load flow calculation (e.g. Excel, Matlab) to calculate $\underline{U}_B$:

Formulas needed

For a network without shunts,

$$Y_{ii} = \sum_{k \neq i} y_{ik}, \quad Y_{ij} = -y_{ij}, \quad y_{ij} = \frac{1}{z_{ij}}.$$

Complete solution

The branch admittances are

$$y_{12} = \frac{1}{j0.25} = -j4 \text{ p.u.}, \quad y_{23} = \frac{1}{j0.1} = -j10 \text{ p.u.}$$

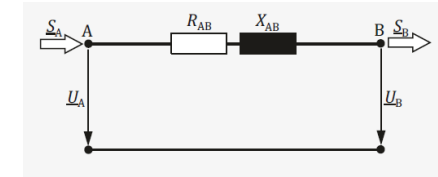
There is no direct branch between buses 1 and 3. Therefore

$$\underline{Y}_{bus} = \begin{bmatrix} y_{12} & -y_{12} & 0 \\ -y_{12} & y_{12} + y_{23} & -y_{23} \\ 0 & -y_{23} & y_{23} \end{bmatrix} = \begin{bmatrix} -j4 & j4 & 0 \\ j4 & -j14 & j10 \\ 0 & j10 & -j10 \end{bmatrix} \text{ p.u.}$$

Every row sums to zero, as expected for a network without shunt admittances.

Exercise 12: Two-node load flow

Exercise



Given:

$$U_A = |U_A| = 400 \text{ V}$$

$$Z_{AB} = 0.1 + j0.1 \, \Omega$$

$$P_B = 9815 \text{ W}$$

$$Q_B = 4908 \text{ Var}$$

To be calculated:

$\underline{U}_B$ (magnitude and phase angle)

Dr. Severin Nowak

Hochschule Technik und Architektur Luzern, 6048 Horw, 14.02.2024

Formulas needed

At load bus B,

$$\underline{S}_B = \underline{U}_B \underline{I}_B^*, \quad \underline{I}_B = \left(\frac{\underline{S}_B}{\underline{U}_B} \right)^*.$$

The fixed-point load-flow equation is

$$\underline{U}_B^{(k+1)} = \underline{U}_A - \underline{Z}_{AB} \left(\frac{\underline{S}_B}{\underline{U}_B^{(k)}} \right)^*.$$

Complete solution

Use

$$\underline{U}_A = 400 + j0 \text{ V}, \quad \underline{Z}_{AB} = 0.1 + j0.1 \, \Omega, \quad \underline{S}_B = 9815 + j4908 \text{ VA},$$

and start with the flat voltage $\underline{U}_B^{(0)} = 400 + j0 \text{ V}$.

First iteration:

$$\underline{I}_B^{(0)} = \left(\frac{9815 + j4908}{400} \right)^* = 24.5375 - j12.2700 \text{ A},$$

$$\begin{aligned} \underline{U}_B^{(1)} &= 400 - (0.1 + j0.1)(24.5375 - j12.2700) \\ &= 396.3193 - j1.2268 \text{ V}. \end{aligned}$$

Continuing:

k	$\underline{U}_B^{(k)} \text{ [V]}$
1	$396.3193 - j1.2268$
2	$396.2813 - j1.2266$
3	$396.2809 - j1.2268$
4	$396.2809 - j1.2268$

The iteration has converged. In polar form,

$$\underline{U}_B = 396.283 \angle (-0.1774^\circ) \text{ V}.$$

The voltage-magnitude drop is $400 - 396.283 = 3.717 \text{ V}$, or ap-

proximately 0.93%.

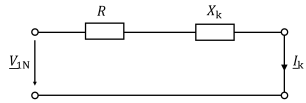
Exercise 13: Short-circuit current

Exercise

Exercices Electrical Power Grids (EPG)

Page 9

✓ EPG-Exercise 13: Short circuits



$V = 20\text{ kV}$ (line voltage)

$f = 50\text{ Hz}$

$I_k = 300\text{ A}$ (Steady state short circuit current)

$R = 12\Omega$

- Determine the value of the leakage reactance X_k .
- Calculate the short-circuit current at time $t = 0.03\text{ s}$. Assume that the short circuit occurred at $t = 0$, i.e., when the voltage crossed zero (rising edge).

$$\begin{aligned}
 i(t) &= i_s(t) + i_t(t) \\
 i_s(t) &= \frac{\sqrt{2}V_s}{|Z|} \sin(\omega t + \alpha - \theta) \\
 i_t(t) &= -i_s(0) \cdot e^{-(R/L)t} = \frac{\sqrt{2}V_s}{|Z|} \sin(\theta - \alpha) \cdot e^{-(R/L)t} \\
 \text{with } |Z| &= \sqrt{R^2 + \omega^2 L^2} \angle \left(\theta = \tan^{-1} \frac{\omega L}{R} \right)
 \end{aligned}$$

The corresponding inductance and impedance angle are

$$L = \frac{36.572}{2\pi 50} = 0.11641\text{ H}, \quad \theta = \tan^{-1} \frac{36.572}{12} = 71.834^\circ.$$

b) Current at $t = 0.03\text{ s}$

The steady-state AC component is

$$\begin{aligned}
 i_s(0.03) &= \sqrt{2}(300) \sin(2\pi 50 \cdot 0.03 - 71.834^\circ) \\
 &= 403.12\text{ A}.
 \end{aligned}$$

The decaying DC component is

$$\begin{aligned}
 i_t(0.03) &= -\sqrt{2}(300) \sin(-71.834^\circ) e^{-(12/0.11641)0.03} \\
 &= 18.30\text{ A}.
 \end{aligned}$$

Therefore the instantaneous short-circuit current is

$$i(0.03) = 403.12 + 18.30 = 421.42\text{ A}.$$

Formulas needed

For one phase of the RL short-circuit equivalent:

$$V_{ph} = \frac{V_{LL}}{\sqrt{3}}, \quad |Z| = \frac{V_{ph}}{I_k}, \quad X_k = \sqrt{|Z|^2 - R^2}, \quad L = \frac{X_k}{\omega}.$$

With voltage zero-crossing angle $\alpha = 0$:

$$i(t) = \frac{\sqrt{2}V_{ph}}{|Z|} \sin(\omega t - \theta) - \frac{\sqrt{2}V_{ph}}{|Z|} \sin(-\theta) e^{-(R/L)t}, \quad \theta = \tan^{-1} \frac{X_k}{R}.$$

Complete solution

a) Leakage reactance

$$V_{ph} = \frac{20\,000}{\sqrt{3}} = 11\,547\text{ V}, \quad |Z| = \frac{11\,547}{300} = 38.490\ \Omega.$$

Thus

$$X_k = \sqrt{38.490^2 - 12^2} = 36.572\ \Omega.$$